Extended evidence accumulation with motor preparation: formal predictions, simulations, and the results that do not follow

Working document

1 Scope and summary

This is a working document, not a draft for the manuscript.

It documents a formal implementation of the extended evidence accumulation theory, derives its predictions, calibrates it against the J Neuroscience experiment, and reports which electromyographic results follow from it and which do not. Two variants are treated in parallel, differing only in when motor preparation begins.

What works. With preparation anchored to stimulus offset, sixteen of the nineteen directional effects come out with the correct sign and are stable across simulation seeds. Premotor execution time is reproduced to within fourteen milliseconds in every condition, premature responses to within one percentage point, and motor execution time is invariant as observed.

What does not. R15, the duration by coherence interaction on delay-period partial-activation rate, comes out with the opposite sign, and R17 fails with it as a corollary. Section 6 shows this is not a calibration failure but a structural incompatibility: R15 and the near-invariance of accuracy with stimulus duration constrain the internal threshold in opposite directions. Nine mechanisms were tested; none removes the contradiction.

A second unresolved point. R3 is reproduced here but does not appear in Achard, Gajdos Preuss & Servant (2025), Experiment 2, which uses a nearly identical design. Section 7 locates the difference precisely and reports that the explanation I first proposed is refuted by the accuracy data.

Convention on statistics. Effects reported as significant are the ANOVA results of the manuscript. Additional analyses with no ANOVA counterpart, in particular subject-level analyses of R15, use planned contrasts and are labelled as such.

2 The empirical results

The labels R1 to R24 are defined once here. They are exhaustive for electromyography: the fifteen significant effects on partial-activation rate in Table 2, the four entries for the two latencies in Table 1, and five statements in the Results and Discussion text. Electroencephalographic measures are excluded.

2.1 Latencies

R1 Premotor execution time, the latency from the response signal to the onset of the muscle activation producing the response, is shorter after a 2-second stimulus.

R2 Premotor execution time is shorter at higher motion coherence.

R3 Duration by coherence interaction on premotor execution time: the coherence effect is larger in the 0.3-second condition.

R4 Motor execution time is unaffected by any factor.

2.2 Channel selectivity

R5 The rate of partial muscle activations is near zero in the channel of the non-selected response.

2.3 Stimulus period, 0 to 2 s after stimulus onset

R6 Main effect of duration.

R7 Main effect of coherence.

R8 Main effect of time bin: the rate ramps upward.

R9 Duration by time interaction: steeper ramp in the 0.3-second condition.

R10 Coherence by time interaction: shallower ramp at lower coherence.

R11 No duration by coherence interaction. This appears only as an absent cell in Table 2 and is never discussed; listed for completeness.

2.4 Delay period, 1 to 3 s after stimulus offset

R12 Main effect of duration: higher rate after a 2-second stimulus.

R13 Main effect of coherence.

R14 Main effect of time bin: continued ramping.

R15 Duration by coherence interaction: the coherence effect is more pronounced after a 2-second stimulus.

R16 Duration by time interaction: the gap between duration conditions grows.

R17 No coherence by time interaction, used as a diagnostic that evidence accumulation has ceased.

2.5 Response-signal period and response period

R18 Main effect of time bin: the rate decreases.

R19 Main effect of duration.

R20 Duration by time interaction.

R21 No coherence effects.

R22 Main effect of time bin in the response period.

R23 Duration by time interaction in the response period.

R24 The rate appears to level off shortly before the response signal in the 2-second, high-coherence condition.

3 Theory

The organising principle of this section is different from the previous version. Rather than developing the stochastic machinery first and applying it afterwards, every subsection below terminates in a statement about one or more of R1–R24, and every such statement carries an explicit status:

Structural true by construction of the model; reproducing it is not evidence for the model.

Theorem follows from the model with no further assumption.

Conditional follows from the model given an explicitly stated property of the accumulation stage, labelled (A1)–(A5), each of which is checkable by direct simulation in one line.

Decomposition the sign is not determined; the effect is written as a sum of terms of known sign, which yields a new testable prediction.

Not predicted outside the reach of the present formalism.

Section 3.14 lists the corrections relative to the previous version of this section. Section 3.15 collects the recomputed constants; several of them differ materially from those previously quoted.

Throughout, the calibrated values are $\lambda = 21.47$, $\kappa = 10.778$, $\sigma_m = 1.8216$, $\alpha = 0.1344$, $g = 1.5897$, $r = 0.6751$, $\theta = 0.8931$, $w = 6.16$, $\sigma_q = 1.213$, $T_{\text{enc}} = 0.22$, all in the units fixed by $\sigma = 1$.

3.1 Standing assumptions

(A0) Model equations. On $[T_{\text{enc}}, T_{\text{stop}}]$,

$$\mathrm{d}x = v_c \mathrm{d}t + \sigma \mathrm{d}W_x, \quad x(T_{\text{enc}}) = 0, \qquad \dot{\tilde{y}} = \lambda(x - \tilde{y}), \quad \tilde{y}(T_{\text{enc}}) = 0, \tag{1}$$

with $v_c = A(c/0.5)^{0.6}$ and $T_{\text{stop}} = \min\{\inf\{t : |\tilde{y}| \geq \theta\}, T_{\text{enc}} + D\}$. Write $T_\theta = \inf\{t : |\tilde{y}| \geq \theta\}$, $s = \operatorname{sign} \tilde{y}(T_{\text{stop}})$, $m_0 = \tilde{y}(T_{\text{stop}})$ and

$$M := |m_0| \in [0, \theta]. \tag{2}$$

For $t > T_{\text{stop}}$ the filter input is $m_0 + \varepsilon$ with $\varepsilon(T_{\text{stop}}) = 0$ and $\mathrm{d}\varepsilon = -\kappa\varepsilon \mathrm{d}t + \sigma_m \mathrm{d}W_\varepsilon$. Preparation is added outside the filter, anchored at $t_a = T_{\text{enc}} + D$ and frozen at $t_R + T_{\text{enc}}$. From $t_R + T_{\text{enc}}$, $\mathrm{d}q = w\mathrm{d}t + \sigma_q \mathrm{d}W_q$ adds $s\, q^+$. The button is pressed when z reaches $g + r$, plus T_r.

(A1) Independence of the maintenance noise. $W_\varepsilon \perp\!\!\!\perp W_x$, and $W_q \perp\!\!\!\perp (W_x, W_\varepsilon)$.

This assumption is doing real work and was previously left implicit. The old text justified $sU \overset{d}{=} U$ "because ε is a symmetric process". That is not sufficient: s is $\mathcal{F}^x_{T_{\text{stop}}}$-measurable, so symmetry of the law of U alone does not licence dropping s — one needs U to be independent of s. Under (A1) it is, and the following holds exactly.

Lemma 1 (Conditional representation)**.** *Under (A0)–(A1), conditionally on (T_{stop}, s, m_0),*

$$z(t) = \mu(t) + V(t - T_{stop}), \qquad \mu(t) = M + \alpha\big(\min(t, t_R + T_{enc}) - t_a\big)^+, \tag{3}$$

where V is a centred Gaussian process, independent of (T_{stop}, s, m_0), with the law of U solving $\dot{U} = \lambda(\varepsilon - U)$, $U(0) = \varepsilon(0) = 0$.

Proof. U is a measurable functional of $\{W_\varepsilon(T_{\text{stop}}+u) - W_\varepsilon(T_{\text{stop}}) : u \geq 0\}$ alone, because ε and U are both reset to 0 at T_{stop}. By (A1) that increment process is independent of $\mathcal{F}^x_{T_{\text{stop}}} \supseteq \sigma(T_{\text{stop}}, s, m_0)$ and is a Brownian motion. Hence $U \perp\!\!\!\perp (T_{\text{stop}}, s, m_0)$, and since the law of U is symmetric, $sU \overset{d}{=} U$ conditionally on s. □

Remark 1. If instead the maintenance perturbation were the continuation of the accumulation noise, Lemma 1 would fail: conditioning on $\{\tilde{y} \text{ hits } +\theta\}$ tilts the recent increments of W_x upward, so sU would start with a positive bias of order $\sigma\sqrt{\lambda/2}\cdot\lambda^{-1}$ decaying over $1/\lambda \approx 47$ ms. The present model avoids this by construction. It should be stated as a modelling choice rather than derived from symmetry.

(A2)–(A5): accumulation-stage properties. Everything below that concerns condition effects is a statement about the law of M and of T_θ. Four properties are used. Let $F_{D,c} = \Pr(T_\theta \le T_{\text{enc}} + D)$.

(A2) *Duration monotonicity.* $F_{D_2,c} \ge F_{D_1,c}$ for $D_2 > D_1$, and $M_{D_1,c} \preceq_{\text{st}} M_{D_2,c}$.

(A3) *Coherence monotonicity.* $F_{D,c_H} \ge F_{D,c_L}$ and $M_{D,c_L} \preceq_{\text{st}} M_{D,c_H}$ for $c_H > c_L$.

(A4) *Hitting-time monotonicity.* $T_\theta^{c_H} \preceq_{\text{st}} T_\theta^{c_L}$.

(A5) *Compression.* The coherence contrast in M is smaller at long duration: $\mathbb{E}M_{2,c_H} - \mathbb{E}M_{2,c_L} \le \mathbb{E}M_{0.3,c_H} - \mathbb{E}M_{0.3,c_L}$.

(A2) is immediate for F because $T_\theta \le T_{\text{enc}} + D_1$ implies $T_\theta \le T_{\text{enc}} + D_2$ pathwise. (A5) is a consequence of saturation at θ and is the property the manuscript invokes for R3. (A3)–(A4) are partly provable:

Lemma 2 (Pathwise ordering of the stopped process)**.** *Couple two coherence levels on the same W_x and write $\hat{y}^c(t) = \tilde{y}^c(t \wedge T_\theta^c)$. Then $\hat{y}^{c_H}(t) \ge \hat{y}^{c_L}(t)$ for all t, almost surely. In particular $m_0^{c_H} \ge m_0^{c_L}$ a.s.*

Proof. Before either process stops, $\tilde{y}^{c_H}(t) - \tilde{y}^{c_L}(t) = \Delta(t) := (v_{c_H} - v_{c_L})\big(t - (1-e^{-\lambda t})/\lambda\big) \ge 0$, since the two solutions of (1) driven by the same W_x differ only by the deterministic response to the drift difference. Suppose H stops first. If it stops at $-\theta$ then $\tilde{y}^{c_L} \le -\theta$ at that instant, so L had already stopped, a contradiction; hence H stops at $+\theta$ and thereafter $\hat{y}^{c_H} = \theta \ge \hat{y}^{c_L}$. Symmetrically, if L stops first it stops at $-\theta$ and thereafter $\hat{y}^{c_L} = -\theta \le \hat{y}^{c_H}$. □

Lemma 3 (Likelihood-ratio ordering of the commitment time)**.** *In the unfiltered limit $\lambda \to \infty$ (threshold applied to x), the density of T_θ satisfies $f_v(t)/f_{v'}(t) = \big[\cosh(v\theta/\sigma^2)/\cosh(v'\theta/\sigma^2)\big]e^{-(v^2-v'^2)t/2\sigma^2}$, which is decreasing in t for $v > v'$. Hence $T_\theta^{c_H} \preceq_{\text{lr}} T_\theta^{c_L}$ and (A4) holds. For finite λ the same Girsanov computation gives $f_v(t)/f_0(t) = \mathbb{E}_0[\cosh(vx_{T_\theta}/\sigma^2) \mid T_\theta = t]\, e^{-v^2t/2\sigma^2}$, and (A4) holds whenever the conditional expectation is non-increasing in t; the departure is controlled by $\text{sd}(x - \tilde{y}) = \sigma/\sqrt{2\lambda} = 0.153$ against $\theta = 0.893$.*

Proof. Girsanov gives $\mathrm{d}P_v/\mathrm{d}P_0|_{\mathcal{F}_{T_\theta}} = \exp(vx_{T_\theta}/\sigma^2 - v^2T_\theta/2\sigma^2)$. Under P_0 the system is invariant under the sign flip $W \mapsto -W$, so conditionally on $T_\theta = t$ the law of x_{T_θ} is symmetric and the exponential may be replaced by its even part. When $\lambda = \infty$, $x_{T_\theta} = \pm\theta$ exactly and the conditional expectation is the constant $\cosh(v\theta/\sigma^2)$. □

Remark 2 (What Lemma 2 does *not* give). The pathwise ordering is of the *signed* m_0; the quantity that drives the muscle is $M = |m_0|$, and $|\cdot|$ destroys the ordering on error trials. The obstruction is confined to sub-threshold error trials, since $M = \theta$ whenever T_θ is reached, irrespective of sign. (A3) is therefore exact whenever every trial reaches θ, and approximate with an error at most $\sup h \cdot \Pr(M < \theta,\ m_0 < 0)$ otherwise. It is checkable in one line of simulation and should be checked rather than assumed.

3.2 The gating lemma: nothing crosses before commitment

This is the single most useful structural fact in the model and it was absent from the previous version.

Lemma 4 (Gating). *Assume $\theta < g$ and offset anchoring. Then for every $t < T_{stop}$, $z(t) < g$ almost surely. Consequently every partial activation, in every analysis window, occurs on a trial that has already committed, and*

$$\nu_{D,c}(t) = \mathbb{E}\Big[\mathbf{1}\{T_{stop} \le t\}\, \rho\big(t - T_{stop};\, g - \mu(t)\big)\Big], \tag{4}$$

where $\rho(\tau; A)$ is the transient crossing rate of Proposition 4.

Proof. Before T_{stop}, $|\tilde{y}| < \theta$ by definition of the first hitting time; under offset anchoring $t_a = T_{\text{enc}} + D \ge T_{\text{stop}}$, so the preparation term is zero on $[0, T_{\text{stop}})$; hence $z = s\tilde{y} \le |\tilde{y}| < \theta < g$. Equation (4) is then Lemma 1 together with the tower property. □

$\theta < g$ is not a calibration accident: $\theta \ge g$ would make every committed trial respond immediately. The lemma therefore holds for the whole admissible parameter region.

Remark 3 (This corrects the previous Proposition 6). The old argument for R9 read: "the level rises at $\bar{v}$ for uncommitted trials and is flat for committed ones". The mean level of uncommitted trials is irrelevant, because uncommitted trials contribute *exactly zero* crossings. The stimulus-period rate is governed entirely by the commitment-time distribution and by what happens after commitment. Every stimulus-period statement below is rewritten accordingly.

3.3 The maintenance process: exact second moments

Lemma 5 (Kernel). $V(\tau) = \int_0^\tau K(\tau - p)\mathrm{d}W(p)$ *with* $K(v) = \dfrac{\lambda\sigma_m}{\lambda - \kappa}\big(e^{-\kappa v} - e^{-\lambda v}\big)$, $K(0) = 0$, $K'(0) = \lambda\sigma_m$.

The proof is unchanged. Two consequences that the previous version did not state explicitly, and which are needed for the stimulus period, follow immediately from $\mathrm{d}/\mathrm{d}\tau \int_0^\tau K^2 = K(\tau)^2$:

Proposition 1 (Exact transient moments). *For all $\tau \ge 0$,*

$$\sigma_V^2(\tau) = \int_0^\tau K^2 = \frac{\lambda^2\sigma_m^2}{(\lambda-\kappa)^2}\left[\frac{1 - e^{-2\kappa\tau}}{2\kappa} - \frac{2(1 - e^{-(\lambda+\kappa)\tau})}{\lambda + \kappa} + \frac{1 - e^{-2\lambda\tau}}{2\lambda}\right], \tag{5}$$

$$\sigma_{\dot{V}}^2(\tau) = \int_0^\tau (K')^2 = \frac{\lambda^2\sigma_m^2}{(\lambda-\kappa)^2}\left[\frac{\lambda(1 - e^{-2\lambda\tau})}{2} - \frac{2\lambda\kappa(1 - e^{-(\lambda+\kappa)\tau})}{\lambda + \kappa} + \frac{\kappa(1 - e^{-2\kappa\tau})}{2}\right], \tag{6}$$

$$\mathrm{Cov}\big(V(\tau), \dot{V}(\tau)\big) = \int_0^\tau KK' = \tfrac{1}{2}K(\tau)^2. \tag{7}$$

As $\tau \to \infty$ these converge to $\sigma_Y^2 = \lambda\sigma_m^2/(2\kappa(\lambda+\kappa))$, $\sigma_{\dot{Y}}^2 = \lambda^2\sigma_m^2/(2(\lambda+\kappa))$ and 0, and $C(\tau) = B[e^{-\kappa|\tau|}/\kappa - e^{-\lambda|\tau|}/\lambda]$ with $B = \lambda^2\sigma_m^2/(2(\lambda-\kappa)(\lambda+\kappa))$; $\beta := \sigma_{\dot{Y}}/\sigma_Y = \sqrt{\lambda\kappa}$.

Identity (7) is the clean form of the statement the old Remark 3 was reaching for. The reduced Rice formula requires $\mathrm{Cov}(V, \dot{V}) = 0$; this is *exactly* zero only in the limit, and the exact transient value is $K(\tau)^2/2$, which decays at rate 2κ. It is what must be carried in the stimulus period, where τ can be a few tens of milliseconds.

Proposition 2 (Rate of convergence, corrected). $\dfrac{\sigma_Y^2 - \sigma_V^2(\tau)}{\sigma_Y^2} = \dfrac{\lambda(\lambda+\kappa)}{(\lambda-\kappa)^2}e^{-2\kappa\tau}\big(1 + o(1)\big) = 6.056\, e^{-2\kappa\tau}\big(1 + o(1)\big)$ *at calibrated parameters.*

The previous version quoted $e^{-2\kappa\tau}$, omitting the prefactor 6.06. At $\tau = 1$ s the deficit is 2.6×10^{-9} rather than 4×10^{-10}; nothing empirical changes, but the stated bound was not correct.

Proposition 3 (Path regularity). *$V \in C^1$ almost surely, and $V \notin C^2$ almost surely. Moreover $C \in C^2(\mathbb{R}) \setminus C^3(\mathbb{R})$.*

Proof. Pathwise: ε is an Ornstein–Uhlenbeck process, hence has continuous paths a.s.; U solves $\dot U = \lambda(\varepsilon - U)$, so $\dot U$ is a continuous function of two continuous functions, hence continuous a.s.; therefore $U \in C^1$. If U'' existed then $\dot\varepsilon = U''/\lambda + \dot U$ would exist, which fails a.s. for an Ornstein–Uhlenbeck process. For C: on $\tau > 0$, $C''''(0^+) = B(\lambda^2 - \kappa^2) \neq 0$, and C''' is odd because C is even, so $C'''(0^-) = -C'''(0^+)$ and $C'''(0)$ does not exist. □

Remark 4 (Why the previous proof of H1 was wrong). The old text asserted that a stationary Gaussian process has a.s. continuously differentiable paths "whenever its covariance is twice differentiable at the origin with C'' continuous". That is false in general: continuity of the covariance of the derivative process does not imply sample continuity of the derivative process, the standard counterexample being $r(0) - r(\tau) \asymp 1/\log(1/|\tau|)$. A sufficient condition here would be $-C''(0) + C''(\tau) = O(|\tau|)$, which does hold, since $-C''(0) + C''(\tau) = B[\lambda(1 - e^{-\lambda|\tau|}) - \kappa(1 - e^{-\kappa|\tau|})] = B(\lambda^2 - \kappa^2)|\tau| + O(\tau^2)$. But the pathwise argument above is shorter and needs no entropy condition, so the general theorem should simply be dropped. The statement "C^2 but not C^4" should become "$C \in C^2 \setminus C^3$", which is strictly stronger and is proved in one line.

3.4 Kac–Rice in the form actually needed

Proposition 4 (General Gaussian crossing rate). *Let $z(t) = \mu(t) + V(\tau)$ with $(V, \dot V)$ jointly Gaussian, $\operatorname{Var} V = s^2 > 0$, $\operatorname{Var} \dot V = \dot s^2 > 0$, $\operatorname{Cov}(V, \dot V) = \gamma$, $\mathbb{E}\dot z = \bar\alpha$. Write $A = g - \mu(t)$, $\eta = \bar\alpha + \gamma A/s^2$, $\omega^2 = \dot s^2 - \gamma^2/s^2$. Then the upcrossing rate of g is*

$$\rho = \frac{1}{s}\varphi\Big(\frac{A}{s}\Big)\Big[\eta\,\Phi(\eta/\omega) + \omega\,\varphi(\eta/\omega)\Big]. \tag{8}$$

Proof. $\nu = p_z(g)\mathbb{E}[\dot z^+ \mid z = g]$. Given $V = A$, $\dot V \sim \mathcal{N}(\gamma A/s^2, \omega^2)$, so $\dot z \mid z = g \sim \mathcal{N}(\eta, \omega^2)$, and $\mathbb{E}[\mathcal{N}(\eta, \omega^2)^+] = \eta\Phi(\eta/\omega) + \omega\varphi(\eta/\omega)$. Hypotheses H1–H3 hold by Proposition 3 and Proposition 1. □

Corollary 1 (Delay-period reduction). *In the delay window $\tau \geq 1$ s, so $\gamma = 0$ to within 10^{-9}, $s = \sigma_Y$, $\dot s = \sigma_{\dot Y}$, $\bar\alpha = \alpha$, and*

$$\nu(t \mid M) = \Gamma_0\, h\big(\mu(t)\big), \qquad h(u) = \frac{1}{\sigma_Y}\varphi\Big(\frac{g-u}{\sigma_Y}\Big), \qquad \Gamma_0 = \alpha\Phi(m) + \sigma_{\dot Y}\varphi(m),\ m = \frac{\alpha}{\sigma_{\dot Y}}. \tag{9}$$

$\Gamma_0 = 2.0107$ and $\nu_{\max} = \Gamma_0\varphi(0)/\sigma_Y = \Gamma\beta/2\pi = 2.506$ per second, where $\Gamma = \sqrt{2\pi}[m\Phi(m) + \varphi(m)] = 1.0350$.

Remark 5 (Corrected bound). The old text bounded the rate by $\beta/2\pi = 2.421$. With a trend the correct bound is $\Gamma\beta/2\pi = 2.506$. The distinction is 3.5 per cent and matters only for absolute rates, but the R24 argument quotes the bound explicitly.

Write $a(t) = (g - \mu(t))/\sigma_Y$ and $k = \alpha/\sigma_Y = 0.4198\ \mathrm{s}^{-1}$. The derivatives of h, used repeatedly below, are

$$h' = \frac{a}{\sigma_Y}h > 0, \qquad h'' = \frac{a^2 - 1}{\sigma_Y^2}h, \qquad h''' = \frac{a(a^2 - 3)}{\sigma_Y^3}h. \tag{10}$$

So h is increasing on $u < g$; convex where $a > 1$; and h' is convex where $a > \sqrt{3}$. These three thresholds do all the work in Sections 3.7 and 3.12, and the calibrated a straddles $\sqrt{3}$ (Section 3.15).

3.5 The three corrections

3.5.1 Trend

Unchanged: Γ is a level-independent multiplicative factor of 1.0350 that cancels from every ratio of rates at the same time point and from every log-slope, and does not cancel from absolute rates.

3.5.2 Absorption

The observed process is killed at $g + r$. The previous version asserted $\nu^{\text{obs}} \le \nu$ on the ground that "killing removes precisely the paths that have been highest". That reasoning is incomplete, and in one respect points the wrong way:

Remark 6 (Two opposing effects). Conditioning on survival lowers the density of $z(t)$ at g. But it also biases the slope *upward*, because $\operatorname{Cov}(V(u), \dot{V}(t)) = C'(t-u) = B[-e^{-\kappa(t-u)} + e^{-\lambda(t-u)}] < 0$ for $u < t$: the process is mean-reverting, so a path constrained to have been low is more likely to be rising now. The net sign is not determined by inspection.

Proposition 5 (A bound that is actually proved)**.** *With* $\tau_{abs} = \inf\{t : z(t) \ge g + r\}$ *and* $p(t) = \Pr(\tau_{abs} \le t)$*,*

$$\nu^{obs}(t) \le \frac{\nu(t)}{1 - p(t)}. \tag{11}$$

Proof. $\mathbb{E}[N_g(t, t+h)\mathbf{1}\{\tau_{\text{abs}} > t\}] \le \mathbb{E}[N_g(t, t+h)]$; divide by h and by $\Pr(\tau_{\text{abs}} > t) = 1 - p(t)$ and let $h \downarrow 0$. □

At the observed premature rate of 1.35 per cent the bound is $\nu^{\text{obs}} \le 1.014\nu$. This is weaker in form than the old claim but it is a theorem, and it is of the size the simulation reports. **However**, the size of the correction quoted previously was based on $a \in [2.2, 3.1]$, which is wrong (Section 3.15): at $a \in [1.4, 1.8]$ the note's own absorption table gives a correction near 1 per cent, not below 0.2 per cent.

3.5.3 Discretisation

Proposition 6 (Euler bias is not multiplicative across conditions)**.** *The Euler scheme with step* Δ *applied to* (ε, U) *has stationary variance*

$$\sigma^2_{Y,\Delta} = \sigma^2_Y\big(1 + \delta\big) + O(\Delta^2), \qquad \delta = \frac{\lambda\kappa}{\lambda + \kappa}\Delta = 7.176\,\Delta. \tag{12}$$

Consequently

$$\ln \nu_\Delta - \ln \nu = \frac{(a^2 - 1)\delta}{2} + \textit{(terms independent of } a) + O(\Delta^2), \tag{13}$$

so the relative bias differs between two conditions with distances a_1, a_2 *by* $\exp[(a_1^2 - a_2^2)\delta/2]$*.*

Proof. Let $A = 1 - \kappa\Delta$, $\rho = 1 - \lambda\Delta$, $b = \lambda\Delta$. Stationarity of the discrete chain gives $\operatorname{Var}\varepsilon = \sigma_m^2\Delta/(1 - A^2)$, $\operatorname{Cov}(U, \varepsilon) = bA \operatorname{Var}\varepsilon/(1 - \rho A)$ and $\operatorname{Var} U(1 - \rho^2) = b^2 \operatorname{Var}\varepsilon + 2\rho b \operatorname{Cov}(U, \varepsilon)$. Expanding to first order in Δ gives $\operatorname{Var}\varepsilon = (\sigma_m^2/2\kappa)(1 + \kappa\Delta/2)$ and $\operatorname{Var} U = \frac{\lambda}{\lambda+\kappa} \operatorname{Var}\varepsilon\big[1 + \frac{\kappa(\lambda-\kappa)}{2(\lambda+\kappa)}\Delta\big]$, whose product is the stated δ. The second display follows from $\ln \nu = \text{const} - \ln \sigma_Y - a^2/2$ with $a^2 \mapsto a^2/(1 + \delta)$. □

At $\Delta = 2$ ms, $\delta = 0.0144$; across the observed spread $a \in [1.45, 1.74]$ the differential bias is 0.6 per cent. That is small in absolute terms but it is of the order of thirty per cent of R16 and comparable with several seed standard deviations. The statement in the previous version — "a multiplicative bias of this kind largely cancels" — is therefore an empirical observation about the particular table, not a property of the estimator.

Recommendation. (ε, U) is a two-dimensional linear SDE with an exact Gaussian transition kernel. Sampling it exactly removes δ entirely; detecting crossings by quadratic interpolation between samples removes most of the residual undercount. The convergence table in the previous version is not consistent with a single $O(\Delta)$ source (the residual falls by less than a factor two when Δ falls by a factor eight), which is a further reason to eliminate the leading term analytically rather than extrapolate it numerically.

3.6 The two master formulas

Everything about R6–R24 is a corollary of these two equations.

Theorem 1 (Master formulas)**.** *Under (A0)–(A1) and Lemma 4:*

$$\textit{delay window } (\tau \geq 1 \text{ s}): \quad \boxed{\nu_{D,c}(t) = \Gamma_0 \, \mathbb{E}\big[h\big(M_{D,c} + \alpha(t - t_a)\big)\big]} \tag{14}$$

$$\textit{stimulus window}: \quad \boxed{\nu_{D,c}(t) = \mathbb{E}\Big[\mathbf{1}\{T_{stop} \leq t\}\, \rho\big(t - T_{stop};\; g - M - \alpha(t - t_a)^+\big)\Big]} \tag{15}$$

with h as in (9) and ρ as in (8).

Equation (14) says that *every* delay-period effect is a functional of the distribution of the maintained level M alone. Equation (15) says that every stimulus-period effect is a functional of the joint law of (T_{stop}, M) alone — the commitment-time distribution convolved with a fixed post-commitment response kernel.

For the stimulus period a further simplification applies. In the 2-second condition $t_a = 2.22$ s lies outside the analysis window, so preparation contributes nothing and $M = \theta$ for every trial that contributes at all:

$$\nu_{2,c}(t) = \int_0^t f^c_{T_\theta}(u)\, \rho\big(t - u;\, g - \theta\big)\mathrm{d}u. \tag{16}$$

In the 0.3-second condition $t_a = 0.52$ s lies *inside* the window and every trial has committed by then, at $M = \theta$ if $T_\theta \leq 0.52$ and at $M = |\tilde{y}(0.52)| < \theta$ otherwise.

3.7 Predictions: delay period (R12–R17, R24)

Throughout this subsection $a(t) > 1$, which holds at calibrated parameters (Section 3.15), so h is increasing and convex on the relevant range.

Theorem 2 (R14: the delay rate ramps — unconditional)**.** *If $M + \alpha(t - t_a) < g$ a.s. then $\partial_t \nu_{D,c} = \Gamma_0 \alpha \mathbb{E}[h'] > 0$.* Status: Theorem. *At calibrated parameters $M + 3\alpha \leq \theta + 0.403 = 1.296 < g = 1.590$, so the hypothesis holds with margin.*

Theorem 3 (R12, R13: main effects of duration and coherence)**.** *If $M_{D_1,c} \preceq_{\text{st}} M_{D_2,c}$ then $\nu_{D_1,c}(t) \leq \nu_{D_2,c}(t)$ at every t (R12); if $M_{D,c_L} \preceq_{\text{st}} M_{D,c_H}$ then $\nu_{D,c_L}(t) \leq \nu_{D,c_H}(t)$ (R13).* Status: Conditional on (A2), (A3). *Only monotonicity of h is used, hence only $\mu < g$ is required. Moreover, decomposing on $\{T_\theta \leq T_{enc} + D\}$,*

$$\nu_{D,c} = \Gamma_0 \Big[F_{D,c} h(\theta + \alpha\tau) + (1 - F_{D,c}) \mathbb{E}\big[h(M + \alpha\tau) \mid M < \theta\big]\Big], \tag{17}$$

and because h is convex the first term dominates: the condition effects are carried mainly by ΔF, not by the sub-threshold level.

Theorem 4 (R16, and the proportionality law)**.** *Under the same hypotheses, $\partial_t\big[\nu_{D_2,c} - \nu_{D_1,c}\big] = \Gamma_0 \alpha \big(\mathbb{E}[h'(M_{D_2} + \alpha\tau)] - \mathbb{E}[h'(M_{D_1} + \alpha\tau)]\big) > 0$ whenever h' is increasing, i.e. whenever $a > 1$. Hence the duration gap grows: R16.* Status: Conditional on (A2). *More sharply, since $h'/h = a/\sigma_Y$,*

$$\boxed{\frac{\partial_t\big[\nu^{c_H} - \nu^{c_L}\big]}{\nu^{c_H} - \nu^{c_L}} \in \Big[k\, a_{\min},\; k\, a_{\max}\Big] = [0.59,\, 0.76]\ \text{s}^{-1}} \tag{18}$$

and identically for the duration contrast: ***any*** *condition effect on the delay-period rate that acts through the maintained level must grow at approximately* 0.6–0.75 *per second, i.e. roughly triple across the two-second analysis window.*

Corollary 2 (R17 is not a saturation null — it contradicts R13)**.** *R16 and R13 are both observed. Equation* (18) *applies to the coherence contrast exactly as it applies to the duration contrast. R17, the absence of a coherence × time interaction, therefore asserts that the coherence contrast does* not *grow, while R13 asserts that it exists. These are jointly incompatible with any account in which coherence acts on the delay-period rate through the maintained level — including but not restricted to the present model.* Status: THEOREM (a constraint on the data, not on the model).

This is a reclassification of R17. The previous version treated R17 as a corollary of R15 and explained it by saturation. Saturation would make R13 vanish along with R17; R13 does not vanish. The manuscript uses R14 together with R17 as its main argument against the threshold hypothesis; the argument should be restated, because R17 as it stands is evidence against a level account of R13 rather than evidence about accumulation.

Theorem 5 (R24: shape of the saturation)**.** *Let* $\langle\cdot\rangle_h$ *denote the h-weighted average over trials. Then*

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\ln\nu = k\langle a\rangle_h, \qquad \boxed{\frac{\mathrm{d}^2}{\mathrm{d}\tau^2}\ln\nu = k^2\big(\mathrm{Var}_h(a)-1\big) = \frac{\alpha^2}{\sigma_Y^4}\mathrm{Var}_h(M) - \frac{\alpha^2}{\sigma_Y^2}} \tag{19}$$

so the delay log-rate is concave if and only if the between-trial standard deviation of the maintained level is smaller than the within-trial maintenance noise: $\mathrm{sd}_h(M) < \sigma_Y$*. Also* $\partial\nu/\partial\mu = (a/\sigma_Y)\nu \to 0$ *as* $\mu \to g$*,* $\partial^2\nu/\partial\mu^2 = \nu(a^2-1)/\sigma_Y^2$ *(convex for* $a > 1$*, concave for* $a < 1$*), and* $\nu \le \Gamma\beta/2\pi$*.* Status: THEOREM.

Equation (19) strictly generalises the previous Corollary 3, which is the degenerate case $\mathrm{Var}_h = 0$ and gives $-k^2$. The generalisation matters: it converts the observed curvature into a measurement of the trial-to-trial spread of the maintained level (Section 3.12), which is the quantity the note elsewhere identifies as the model's largest deficiency.

Theorem 6 (R15: necessary condition)**.** *Because h is increasing and* $M \le \theta$*,* $\Gamma_0 F_{D,c} h(\theta+\alpha\tau) \le \nu_{D,c} \le \Gamma_0 h(\theta+\alpha\tau)$*, so*

$$\nu_{D,c_H} - \nu_{D,c_L} \le \Gamma_0 h(\theta+\alpha\tau)\big(1 - F_{D,c_L}\big). \tag{20}$$

Integrating over the delay window and writing $N_\theta = \Gamma_0 \int h(\theta+\alpha\tau)\mathrm{d}\tau = 2.097$*,*

$$\boxed{1 - F_{2,c_L} \;\ge\; \frac{\Delta_c N_2}{N_\theta}} \tag{21}$$

With the observed $\Delta_c N_2 = 1.894 - 1.682 = 0.212$ *this gives* $1 - F_{2,0.12} \ge 0.101$*:* ***at least ten per cent of two-second, twelve-per-cent-coherence trials must fail to reach the internal threshold.*** Status: THEOREM.

This replaces the qualitative "requires a substantial fraction" of the previous Corollary 4 with a number. It is necessary, not sufficient; the θ-sweep of Section 6 shows the sufficient requirement is considerably stronger. The logical structure of the R15 argument should be: prove the bound, then use the accuracy data to contradict it, and demote the θ-sweep to an illustration.

Theorem 7 (R3 and R15 are compatible only inside a narrow window)**.** *Write* $\Delta_c\mu_D$ *for the coherence contrast in the level at the response signal. To first order, premotor execution time satisfies* $\partial\mathbb{E}[PMET]/\partial\mu = -1/w$ *(Proposition 10), while the delay rate satisfies* $\partial\nu/\partial\mu = h'\Gamma_0$*. Hence*

$$R3 \iff \Delta_c\mu_{0.3}/\Delta_c\mu_2 > 1, \tag{22}$$

$$R15 \iff \Delta_c\mu_{0.3}/\Delta_c\mu_2 < r^* := \frac{h'(\mu_2)}{h'(\mu_{0.3})} = \frac{a_2}{a_{0.3}}\cdot\frac{\nu_2}{\nu_{0.3}}. \tag{23}$$

At the observed rates and distances $r^* = 1.31$. Status: THEOREM, given the first-order PMET map.

The empirical content is sharp. In J Neuroscience the PMET coherence contrasts are 28.2 ms (short) against 7.1 ms (long), a ratio of 3.97, far outside $(1, 1.31)$: R3 and R15 are incompatible in that data set, and no calibration can reconcile them. In AGPS Experiment 2 the same contrasts are 30.5 and 27.2 ms, a ratio of 1.12, which is *inside* $(1, 1.31)$ — and AGPS is the experiment in which R15 is larger and R3 is absent. The two experiments differ precisely in whether the compression ratio exceeds r^*. This is a considerably more informative statement than "R3 is reproduced here and does not replicate there".

3.8 Predictions: stimulus period (R6–R11)

All of these follow from (15). The previous version treated them as simulation results with one heuristic proposition; the gating lemma makes them tractable.

Theorem 8 (R8: the stimulus-period rate ramps)**.** *$\nu_{D,c}(t)$ is increasing in t on the analysis window, because (i) $\Pr(T_{stop} \le t)$ is non-decreasing; (ii) $\rho(\tau; A)$ is increasing in τ wherever $A > \sigma_V(\tau)$, since $\partial_s[s^{-1}\varphi(A/s)] = s^{-2}\varphi(A/s)(A^2/s^2 - 1)$, and the covariance term $\gamma = K(\tau)^2/2$ adds a further positive conditional drift $\gamma A/s^2$; (iii) in the 0.3-second condition preparation adds $\alpha(t-0.52)^+$ from inside the window.* Status: THEOREM.

Proposition 7 (R6, R9: exact decomposition)**.** *Couple the two duration conditions on the same W_x; they are identical up to 0.52 s. For $t > 0.52$,*

$$\nu_{0.3,c}(t) - \nu_{2,c}(t) = \underbrace{\mathbb{E}\big[\mathbf{1}\{T_\theta \le 0.52\}\big(\rho(t - T_\theta; g - \theta - \alpha(t - 0.52)) - \rho(t - T_\theta; g - \theta)\big)\big]}_{\text{(i) preparation, } \ge 0} \tag{24}$$

$$+\underbrace{\mathbb{E}\big[\mathbf{1}\{T_\theta > 0.52\}\rho(t - 0.52;\, g - |\tilde{y}(0.52)| - \alpha(t - 0.52))\big]}_{\text{(ii) sub-threshold committers, } \ge 0} - \underbrace{\mathbb{E}\big[\mathbf{1}\{0.52 < T_\theta \le t\}\rho(t - T_\theta; g - \theta)\big]}_{\text{(iii) late threshold-reachers, } \ge 0}. \tag{25}$$

R6 is the statement that (i)+(ii)>(iii) integrated over the window; R9 is the same statement for the time derivative. Term (i) grows multiplicatively at rate $k\,a$; term (iii) grows at the rate of the commitment hazard after 0.52 s. Status: DECOMPOSITION.

This supersedes the old Proposition 6. The mechanism is not that "the level rises at $\bar{v}$ for uncommitted trials"; uncommitted trials contribute nothing. R9 is the competition between the preparation ramp in the short condition and the continuing commitment hazard in the long one. It yields a directly testable consequence: R9 should be carried by the post-0.52 s portion of the window and should be small or absent at high coherence, where the commitment hazard in the 2-second condition is already exhausted by 0.52 s.

Theorem 9 (R7: main effect of coherence)**.** *If $T_\theta^{c_H} \preceq_{\text{st}} T_\theta^{c_L}$ (A4) and $\tau \mapsto \rho(\tau; g - \theta)$ is non-decreasing (Theorem 8(ii)), then (16) gives $\nu_{2,c_H}(t) \ge \nu_{2,c_L}(t)$ pointwise. In the 0.3-second condition the same argument applies to the θ-reaching component and (A3) to the remainder.* Status: CONDITIONAL on (A3), (A4); (A4) proved in Lemma 3 in the unfiltered limit.

Proposition 8 (R10: two components of opposite sign — a new prediction)**.** *In the 0.3-second condition the ramp after 0.52 s is multiplicative, so its absolute slope is proportional to the rate, which is larger at high coherence: this produces R10 with the observed sign. In the 2-second condition the rise is driven by $F_c(t)$, which saturates* earlier *at high coherence, producing the opposite sign.* Status: DECOMPOSITION, with the prediction:

> R10 should be positive when computed within the 0.3-second condition alone, and zero or negative when computed within the 2-second condition alone.

The pooled effect is observed to be small (+0.00041) and the model's is small (+0.00172), consistent with near-cancellation, but the split has never been examined.

Proposition 9 (R11)**.** *Nothing in (15) forces the duration × coherence interaction to vanish; the model predicts a small nonzero value whose sign depends on how fast $F_{2,c}$ saturates relative to the sub-threshold spread at 0.3 s.* Status: NOT SIGNED; predicted small.

3.9 Predictions: latencies and channel selectivity (R1–R5)

Proposition 10 (The PMET map)**.** *Let $\mu_R = M + \alpha\,(t_R + T_{enc} - t_a)$ be the level at the moment the second accumulation begins. Ignoring the maintenance fluctuation, the onset of the response burst is the first passage of g by $\mu_R + q$ with $\mathrm{d}q = w\mathrm{d}t + \sigma_q \mathrm{d}W_q$, so*

$$\mathbb{E}[PMET \mid \mu_R] = T_{enc} + \frac{g - \mu_R}{w}, \qquad \frac{\partial \mathbb{E}[PMET]}{\partial \mu_R} = -\frac{1}{w}, \tag{26}$$

the first identity being exact for a one-sided barrier with positive drift. The maintenance fluctuation V shortens this by a term independent of (D, c) to first order.

Theorem 10 (R1, R2)**.** *Under (A2), (A3), μ_R is stochastically larger after a longer stimulus and at higher coherence, so premotor execution time is shorter in both.* Status: CONDITIONAL on (A2), (A3).

Proposition 11 (R3)**.** *$\Delta_c \mathbb{E}[PMET]_D = -\Delta_c \mathbb{E}[\mu_R]_D / w$, so R3 is exactly (A5). The model does not determine the sign of (A5); the Methods of the manuscript say so explicitly. R3 should therefore be reported as evidence against the temporal-preparation hypothesis and not as evidence for the combined hypothesis.* Status: NOT PREDICTED (sign).

Proposition 12 (R4: motor execution time is invariant by construction, to first order)**.** *Motor execution time is the first passage from g to $g+r$ plus T_r. Conditionally on being at g, the subsequent dynamics are driven by q and V, whose laws contain no (D, c)-dependent parameter, so $T_{motor} \perp (D, c)$ exactly at first order. A second-order dependence survives, because the* decomposition *of g into $\mu + q^+ + V$ at onset differs across conditions: a trial arriving at g with a large mean-reverting V and a small q is likelier to fall back. The model therefore predicts a small residual, which is what is observed (2.5 ms on 160 ms, 1.6 per cent).* Status: STRUCTURAL at first order. *Reproducing R4 in simulation is not evidence for the model and should not be counted as such.*

Proposition 13 (R5)**.** *The model represents only the selected channel; the non-selected channel carries $-s\,y$, which never approaches g.* Status: STRUCTURAL. Not a test.

3.10 Predictions: response-signal and response periods (R18–R23)

The previous version had no theory here. The key observation is that the second accumulation does not begin until $t_R + T_{\mathrm{enc}} = 0.22$ s, which is *after* the 0–0.2 s response-signal window. Inside that window the process is still the delay process. What changes is the risk set.

Theorem 11 (R18: attrition, not dynamics)**.** *Let $S(t) = \Pr(PMET > t)$ and let $\eta(t) = -\mathrm{d} \ln S / \mathrm{d}t$ be the hazard of response onset. Since a trial contributes no further partial activations once its response burst has begun, the per-trial observed rate is $\nu^{obs}(t) = \nu^{free}(t) S(t)$ and*

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t} \ln \nu^{obs} = k\langle a \rangle_h - \eta(t)} \tag{27}$$

R18 (a decrease) holds if and only if $\eta(t) > k\langle a \rangle_h \approx 0.7\ s^{-1}$. With the observed PMET distribution (mean ≈ 350 ms, sd ≈ 180 ms), η rises from 0.35 s^{-1} at $t = 0$ to 1.96 s^{-1} at $t = 0.2$ s, so R18 holds over most of the window. Status: THEOREM, parameter-free given the observed PMET distribution.

Two consequences, both new and both testable in the existing data:

1. The decline should *accelerate* across the window and should be flat or slightly positive in the first bin, where $\eta < k\langle a \rangle_h$.

2. Dividing the observed response-signal-locked rate by the empirical survival function $\hat{S}(t)$ should restore a rising profile continuous with the delay period. If it does not, the attrition account fails.

This also explains why the implementation error described in Section 4 inverted R18: that error changed exactly the treatment of the terminal burst, which is the attrition term.

Proposition 14 (R19)**.** *At $t = 0$ the ratio $\nu_{2,c}/\nu_{0.3,c}$ equals its delay-period value (continuity), and the two survival functions differ only by the PMET difference of 50 ms. R19 therefore follows from R12 with a small correction.* Status: CONDITIONAL on (A2).

Proposition 15 (R20)**.** $\mathrm{d}\ln \nu_2^{obs}/\mathrm{d}t - \mathrm{d}\ln \nu_{0.3}^{obs}/\mathrm{d}t = -[\eta_2(t) - \eta_1(t)] < 0$ *because premotor execution time is shorter after a 2-second stimulus (R1), so its hazard is uniformly higher.* Status: THEOREM, given R1. *Note the linkage: the sign of R20 is forced by the sign of R1. At the observed PMET means the hazard difference at $t = 0.15$ s is 0.52 s^{-1}.*

Proposition 16 (R21 is a power null, not a structural zero)**.** *The coherence contrast in this window equals the delay coherence contrast (R13) modulated by the coherence difference in hazard (from R2), which acts in the opposite direction. The predicted magnitude is $\approx$ (R19/R12) $\times$ R13 ≈ 0.012 per bin, about one third of R19.* Status: DECOMPOSITION; the model does not predict an exact zero and the observed null is consistent with insufficient power.

Theorem 12 (R22: the response-locked rate must vanish at the response)**.** *Align at the response onset $t = 0$. A counted partial activation at $t < 0$ is an upcrossing of g that is followed by a downcrossing before 0, since the excursion that reaches $g + r$ is the response and is not counted. Hence*

$$\nu^{resp}(t) = \nu(t)\,\Pr(\textit{excursion beginning at } t \textit{ ends before } 0) \longrightarrow 0 \quad \textit{as } t \uparrow 0, \tag{28}$$

with a decay scale equal to the mean excursion duration. R22 is therefore structural, and its scale *is predicted: $\bar{D} = \Phi(-a)/\nu \approx 78$ ms at calibrated parameters.* Status: THEOREM.

Proposition 17 (R23)**.** *The decline of Theorem 12 is multiplicative and common to conditions, so an absolute-rate interaction of sign $-\mathrm{sign}(\nu_2 - \nu_{0.3})$ follows immediately from R19.* Status: THEOREM, given R19.

3.11 Excursion duration

Proposition 18 (Duration identity)**.** $\dfrac{\mathbb{E}A[0,T]}{\mathbb{E}N[0,T]} = \dfrac{\int_0^T \Phi(-a(t))\mathrm{d}t}{\int_0^T \nu(t)\mathrm{d}t}.$

The proof is unchanged. Two clarifications:

1. The ratio of expectations is not the expectation of a ratio, but it *is* the estimand of the pooled empirical estimator (total time above threshold divided by total number of bursts, pooled over trials), which is how the quantity is measured. The previous version's interpretation is therefore correct for the intended use, and the critique that $\mathbb{E}A/\mathbb{E}N$ is not "the mean duration of a randomly selected excursion" does not bite.

2. The boundary correction is $O(1/\mathbb{E}N)$ relative, and $\mathbb{E}N \approx 1$ per trial in the delay window, so it is not asymptotically negligible: the numerical check gives 2 per cent. The phrase "up to the contribution of at most one incomplete excursion at each endpoint" should be accompanied by that number.

3.12 Quantitative diagnostics

These are new. Each is a number that the theory extracts from the published summary statistics without any simulation.

D1. The standardised distance to the gate. From $\nu = \nu_{\max} e^{-a^2/2}$ with $\nu_{\max} = 2.506$ s^{-1} and the observed delay-window totals (per trial, over the 2 s window): $a = 1.74$ (0.3 s, pooled), $a = 1.46$ (2 s, pooled), $a = 1.40$ (2 s, $c = 0.50$). Two independent checks agree: (i) the observed log-slope of the delay rate, $+0.809$ then $+0.702$ s^{-1}, gives $a = \text{slope}/k = 1.93$ then 1.67; (ii) the premature-response rate of 1.35 per cent requires $\nu(g + r) \approx 0.003$ s^{-1}, i.e. $a + \rho_r \approx 3.7$ with $\rho_r = 2.11$, hence $a \approx 1.6$. **The value $a \in [2.2, 3.1]$ quoted in the previous version is not consistent with the note's own rates**; at $a = 2.5$ the predicted premature rate is 0.02 per cent against 1.35 observed. Consequences: the absorption correction is about 1 per cent rather than below 0.2; and a straddles $\sqrt{3} = 1.732$, which is the sign-change point of h'''.

D2. The trial-to-trial spread of the maintained level, measured from curvature. Equation (19) with the observed change in log-slope, -0.107 s^{-2}, and k estimated from the slope itself gives $\text{Var}_h(M)/\sigma_Y^2 = 1 - 0.107/k^2$, i.e. $\text{sd}_h(M) \approx 0.24$ in units of $\sigma = 1$, about $0.75\sigma_Y$. This is an EMG-derived estimate of exactly the quantity the model is suspected of underestimating. It should be compared with the simulated $\text{sd}(M)$, which is small because most trials saturate at θ.

D3. The premotor-execution-time / rate consistency test. Both PMET and the delay rate are monotone functions of the same maintained level, so each implies a value of $\Delta\mu$. Using $\partial\text{PMET}/\partial\mu = -1/w$ and $\Delta \ln \nu = a\Delta\mu/\sigma_Y$:

contrast	$\Delta\mu$ from PMET	$\Delta\mu$ from delay rate	ratio
duration (pooled)	0.308	0.090	3.4
coherence 12 → 50, 0.3 s	0.174	0.009	19.8
coherence 12 → 50, 2 s	0.044	0.026	1.7

The two measures are quantitatively irreconcilable in the 0.3-second condition, by a factor of twenty. The data require the maintained level to be nearly identical across coherence after a short stimulus (the delay-rate contrast is 5 per cent) while premotor execution time differs by 28 ms. This is a stronger and more localised statement than "the model is uniformly too reactive to condition", and it identifies where the model must be changed.

D4. What D3 implies for the model. A quantity that shifts premotor execution time without shifting the delay-period level is required. The natural candidate is already named in the manuscript: a condition-dependent *starting point of the second accumulation*, distinct from the maintained motor level and added only at $t_R + T_{\text{enc}}$. Such a term

- shifts PMET (R1–R3) by $-\Delta q_0/w$;
- leaves the delay-period rate untouched, hence leaves R12–R17 to be carried by the small level differences the EMG data actually show;
- leaves motor execution time invariant (R4), because it acts before the burst onset and MET is measured from g;
- predicts that the duration and coherence contrasts on the partial-activation rate should *step up* only after $T_{\text{enc}} = 0.22$ s post-signal, i.e. outside the 0–0.2 s response-signal window and inside the response-locked window.

The last item is a sharp falsifiable prediction that distinguishes this variant from the present one. It has not been tested.

D5. The d' benchmark, computed for the actual process. For (1) without stopping, $\mathbb{E}\tilde{y}(D) = v(D - (1 - e^{-\lambda D})/\lambda)$ and $\operatorname{Var}\tilde{y}(D) = \sigma^2[D - 2(1 - e^{-\lambda D})/\lambda + (1 - e^{-2\lambda D})/(2\lambda)]$, so the d' ratios predicted by unbounded accumulation *in this model* are 1.871 (1000 vs 300 ms) and 1.422 (2000 vs 1000 ms), against 1.826 and 1.414 for the unfiltered diffusion and 1.26 and 1.01 observed. The filter makes the prediction slightly *larger*, so the argument that accumulation terminates is strengthened, not weakened, by using the correct process. The previous version should either quote these numbers or label the $\sqrt{T}$ figures explicitly as a diffusion benchmark.

D6. Jensen: the variance conjecture points the wrong way for $a > \sqrt{3}$. Expanding (14) to second order, $\nu \approx \Gamma_0 h(\bar{m})[1 + (a^2 - 1)\operatorname{Var}(M)/2\sigma_Y^2]$, so a condition contrast at common variance V obeys $\Delta\nu \approx \Gamma_0\left[h'(\bar{m}) + \frac{V}{2}h'''(\bar{m})\right]\Delta\bar{m}$. By (10), $h''' > 0$ for $a > \sqrt{3}$ and $h''' < 0$ for $a < \sqrt{3}$. Hence:

> *Where $a > \sqrt{3}$, larger trial-to-trial variance in the maintained level* ***amplifies*** *condition effects; where $a < \sqrt{3}$ it attenuates them.*

The note conjectures that the model's 45 per cent underestimate of premotor-execution-time variability explains its overestimate of seven condition effects. At $a > \sqrt{3}$ that conjecture is backwards: too little variance would predict effects that are too *small*. Since the calibrated $a \in [1.40, 1.74]$ straddles $\sqrt{3} = 1.732$, the sign differs across cells, and the conjecture cannot be evaluated by intuition. If the variance deficit does explain the overestimation, it must do so through its effect on the *means* — for example, trial-to-trial drift variability pushes more trials to saturation at θ in both conditions and compresses $\Delta\bar{m}$ — and not through Jensen. That is a different and separately testable mechanism.

3.13 Status of R1–R24

	status	basis
R1	Conditional (A2)	Prop. 10, $\partial\text{PMET}/\partial\mu = -1/w$
R2	Conditional (A3)	same
R3	Not predicted (sign)	equals (A5); manuscript's Methods concur
R4	Structural (1st order)	no (D, c) parameter after g; residual $\approx 1.6\%$
R5	Structural	architecture; not a test
R6	Decomposition	Prop. 7, terms (i)+(ii) vs (iii)
R7	Conditional (A3),(A4)	Thm. 9; (A4) proved unfiltered
R8	Theorem	Thm. 8
R9	Decomposition	Prop. 7; old Prop. 6 mechanism corrected
R10	Decomposition	Prop. 8; new within-duration prediction
R11	Not signed	predicted small
R12	Conditional (A2)	Thm. 3
R13	Conditional (A3)	Thm. 3
R14	**Theorem**	Thm. 2, unconditional
R15	**Theorem** (bound)	Thm. 6: $1 - F_{2,0.12} \geq 0.101$; Thm. 7
R16	Conditional (A2)	Thm. 4, needs only $a > 1$
R17	**Theorem** (against)	Cor. 2: incompatible with R13
R18	**Theorem**	Thm. 11, from observed PMET hazard
R19	Conditional (A2)	continuity with R12
R20	**Theorem** given R1	hazard ordering
R21	Decomposition	predicted ≈ 0.012/bin; power null
R22	**Theorem**	Thm. 12; scale $= \bar{D} \approx 78$ ms
R23	**Theorem** given R19	multiplicative decline
R24	**Theorem**	Thm. 5, incl. curvature identity

Nine results are now theorems of the model or constraints on the data, against one (R24) previously. Four more are conditional on two checkable properties of the accumulation stage. Three are decompositions that generate new tests. The headline "sixteen of nineteen directional effects come out" understates what the model delivers and overstates what the simulation establishes.

3.14 Errata relative to the previous Section 4

1. **H1.** The general Gaussian regularity theorem invoked is false as stated. Replaced by the pathwise argument (Prop. 3).

2. C^2 **but not** C^4 $\rightarrow C \in C^2(\mathbb{R}) \setminus C^3(\mathbb{R})$, because C''' is odd and $C'''(0^+) \neq 0$.

3. $sU \overset{d}{=} U$**.** Justified by independence (A1), not by symmetry of ε; s is measurable with respect to the accumulation filtration.

4. **Transient bound.** The relative variance deficit is $6.056\, e^{-2\kappa\tau}$, not $e^{-2\kappa\tau}$. Also $\mathrm{Cov}(V, \dot{V})(\tau) = K(\tau)^2/2$ exactly, which is what the reduced formula requires to vanish.

5. **Absorption.** $\nu^{\mathrm{obs}} \leq \nu$ is asserted without proof and the two conditioning effects have opposite signs, since $\mathrm{Cov}(V(u), \dot{V}(t)) < 0$ for $u < t$. Replaced by the proved bound $\nu^{\mathrm{obs}} \leq \nu/(1-p)$.

6. **The value of** a**.** $[2.2, 3.1]$ is inconsistent with the note's own rates and premature-response rate; the correct range is $[1.40, 1.74]$. The absorption correction is therefore about 1 per cent, not below 0.2.

7. **Upper bound on the rate.** $\Gamma\beta/2\pi = 2.506$, not $\beta/2\pi = 2.421$.

8. **Discretisation.** The Euler bias is $\delta = \lambda\kappa\Delta/(\lambda+\kappa)$ on σ_Y^2 and enters $\ln\nu$ as $(a^2-1)\delta/2$; it is therefore *not* condition-independent. Use exact transition sampling.

9. **Proposition 6 (R9).** The mechanism is misattributed: pre-commitment trials contribute exactly zero crossings because $\theta < g$. Replaced by Prop. 7.

10. **Duration identity.** The ratio-of-expectations reading is correct for the pooled estimator, but the boundary correction is $O(1/\mathbb{E}N)$ with $\mathbb{E}N \approx 1$ and should be quoted as the measured 2 per cent.

11. d' **benchmark.** Computed for the filtered process the ratios are 1.871 and 1.422, not 1.826 and 1.414.

12. **Corollary on R15.** "Requires a substantial fraction" replaced by $1 - F_{2,c_L} \geq \Delta_c N_2/N_\theta = 0.101$.

13. **The variance conjecture.** At $a > \sqrt{3}$ the Jensen term implies that too little variance produces effects that are too *small*, the opposite of the note's conjecture. The calibrated a straddles $\sqrt{3}$; the conjecture must be evaluated numerically, not asserted.

14. **R17.** Not a corollary of R15 explained by saturation. Given R13 and the proportionality law (18), R17 is incompatible with any level-based account of the coherence effect.

3.15 Recomputed constants

quantity	value	source
σ_Y	0.32014	$\sqrt{\lambda\sigma_m^2/(2\kappa(\lambda+\kappa))}$
$\sigma_{\dot{Y}}$	4.86988	$\sqrt{\lambda^2\sigma_m^2/(2(\lambda+\kappa))}$
$\beta = \sqrt{\lambda\kappa}$	15.212 s^{-1}	
Γ	1.03497	trend factor
Γ_0	2.01074	$\alpha\Phi(m) + \sigma_{\dot{Y}}\varphi(m)$
$\nu_{\max} = \Gamma\beta/2\pi$	2.5057 s^{-1}	
$k = \alpha/\sigma_Y$	0.41982 s^{-1}	
$\rho_r = r/\sigma_Y$	2.1088	
$(g-\theta)/\sigma_Y$	2.1760	a at commitment, saturated trial
a at delay onset / offset (saturated)	1.756 / 0.916	$\mu = \theta + \alpha\tau$, $\tau \in [1,3]$
a inferred, 0.3 s / 2 s / max cell	1.742 / 1.462 / 1.395	from observed totals
N_θ (saturated window total)	2.097	$\Gamma_0 \int_1^3 h(\theta+\alpha\tau)\mathrm{d}\tau$
transient prefactor	6.0564	$\lambda(\lambda+\kappa)/(\lambda-\kappa)^2$
δ/Δ (Euler)	7.1758 s^{-1}	$\lambda\kappa/(\lambda+\kappa)$
r^* (R3/R15 window)	1.314	$(a_2/a_{0.3})(\nu_2/\nu_{0.3})$
d' ratios, filtered	1.871, 1.422	1000/300, 2000/1000 ms

3.16 Open questions the theory does not close

1. (A3) and (A5) are assumed, not proved. Lemma 2 gives the pathwise ordering of the *signed* m_0; the passage to $|m_0|$ fails only on sub-threshold error trials, and the residual is bounded by $\sup h \cdot \Pr(M < \theta,\, m_0 < 0)$. This is a five-line simulation check and should be done.

2. The observed R18 magnitude has not been reconstructed from the observed PMET distribution. Theorem 11 makes that a one-line prediction with no free parameter.

3. The response-stage observable is defined by the burst-detection rule, not by the process. Nothing in Section 3.10 survives a change in how the terminal burst is treated, which is exactly what previously inverted R18. The rule should be stated in the note as part of the model.

4. D3 and D4 identify a discrepancy the present model cannot absorb. Whether a condition-dependent second-accumulation starting point resolves it, without breaking R4 or R19–R23, has not been tested.

4 Method

4.1 Analysis windows

A single module defines the four windows of the manuscript (p. 15, lines 284–289) — the stimulus period 0 to 2 s after onset, the delay 1 to 3 s after offset, the response-signal period 0 to 0.2 s, the response period −0.2 to 0 s — and a single binning function is applied to data and simulation alike. The dependent variable is the mean number of partial activations per 0.1 s bin per trial. Trials with a premature response, no response, an anticipation or a technical problem are excluded, as in the manuscript.

This gives 20678 trials, the number reported. As a check on the extraction, all nineteen reported effects come out with the correct sign, and the range of motor execution time across conditions is 3.0 ms against 2.5 ms reported.

4.2 Calibration and stability

Ten parameters were calibrated by Nelder–Mead against six accuracies, six premotor execution times, six motor execution times, six premature rates, six stimulus-period totals and six delay-period totals. Every reported effect is the mean over six seeds of 3000 trials per cell, and is called unstable when its mean is smaller than twice its standard deviation across seeds.

Remark 7 (Two errors corrected relative to earlier versions). First, the burst in progress when a trial reached $g + r$ was counted as a partial activation, because removing the trial from the risk set triggered a spurious end-of-burst event. This inflated the response-locked windows twenty-fold and inverted R18. Second, earlier evaluations were run at a gate position where the model produced 0.06 partial activations per trial against 2.03 observed, so the sign tests were performed on noise. Both are fixed.

5 Results

5.1 Offset anchoring

Calibrated: $A = 4.234$, $\theta = 0.893$, $g = 1.590$, $r = 0.675$, $\kappa = 10.78$, $\sigma_m = 1.822$, $\alpha = 0.134$, $w = 6.16$, $\sigma_q = 1.213$, $T_r = 0.047$. These imply $\sigma_Y = 0.320$ and $\beta = 15.2$ per second.

Table 1: Simulated / observed, offset anchoring, six seeds of 3000 trials per cell.

cell	accuracy	PMET (ms)	MET (ms)	premature	stimulus	delay
0.3 s, coh 0.00	0.500 / 0.493	393 / 396	153 / 163	0.49 / 0.72	0.213 / 0.331	0.611 / 1.061
0.3 s, coh 0.12	0.754 / 0.714	380 / 384	155 / 160	0.69 / 0.60	0.292 / 0.357	0.810 / 1.093
0.3 s, coh 0.50	0.941 / 0.951	342 / 356	161 / 160	1.46 / 0.97	0.507 / 0.411	1.393 / 1.146
2.0 s, coh 0.00	0.498 / 0.504	328 / 341	164 / 161	1.58 / 1.54	0.204 / 0.196	1.676 / 1.591
2.0 s, coh 0.12	0.931 / 0.789	312 / 326	165 / 162	1.99 / 1.55	0.283 / 0.233	1.888 / 1.682
2.0 s, coh 0.50	0.976 / 0.952	312 / 319	166 / 162	2.01 / 2.72	0.344 / 0.330	1.914 / 1.894

R9 is reproduced almost exactly, +0.00061 against +0.00064. R20 falls below the stability threshold, which is expected given that the response-signal window has two bins.

Remark 8 (A systematic bias in magnitude). The signs are almost all right, but seven effects exceed the observed 95 per cent confidence interval and none falls below it: R6, R7, R8, R10, R12, R13, R16. The model is uniformly too reactive to condition. The same direction of error appears in the trial-to-trial variability of premotor execution time, which the model underestimates by 45 per cent (92–119 ms simulated against 161–214 ms observed, computed per participant so that between-participant differences are already removed). A model that understates trial-to-trial variability mechanically overstates condition effects. This is a single candidate explanation for two distinct anomalies; it has not been tested.

5.2 Commitment anchoring, at its own best fit

Eleven parameters were calibrated for this variant, adding a saturation level C with $\theta < C < g$, because Proposition **??** requires $\alpha > \bar{v}$ and a large α without saturation drives the level past the gate during the delay. Best fit: $A = 3.871$, $\theta = 0.889$, $g = 1.696$, $r = 0.683$, $\kappa = 13.64$, $\sigma_m = 1.765$, $\alpha = 1.198$, $w = 6.36$, $\sigma_q = 1.214$, $T_r = 0.053$, $C = 1.189$. Final loss 39.9 against 6.6 for the offset variant.

The calibration does what Proposition **??** predicts: it raises α from 0.134 to 1.198, a nine-fold increase, until preparation outruns accumulation. R9 and R6 both recover the correct sign. But the fit collapses elsewhere.

Seven of nineteen effects are correct and robust, twelve are indeterminate, and none fails outright — because almost every effect has been reduced to zero.

Table 2: All nineteen effects, offset anchoring.

	observed	simulated	seed SD	verdict
R1	+0.04990	+0.05395	0.00121	OK
R2	+0.03060	+0.03340	0.00167	OK
R3	+0.01790	+0.03494	0.00201	OK
R6	+0.00566	+0.00301	0.00043	OK
R7	+0.00536	+0.01082	0.00064	OK
R8	+0.00097	+0.00170	0.00003	OK
R9	+0.00064	+0.00061	0.00009	OK
R10	+0.00041	+0.00172	0.00018	OK
R12	+0.03115	+0.04438	0.00037	OK
R13	+0.00970	+0.02550	0.00062	OK
R14	+0.00551	+0.00390	0.00008	OK
R15	+0.21040	−0.02723	0.00248	**FAIL**
R16	+0.00139	+0.00230	0.00012	OK
R17	−0.00032	+0.00207	0.00021	**FAIL**
R18	−0.03602	−0.02408	0.00442	OK
R19	+0.03987	+0.05180	0.00320	OK
R20	−0.01900	−0.01449	0.00748	unstable
R22	−0.02459	−0.02543	0.00172	OK
R23	−0.00556	−0.00454	0.00155	OK

Table 3: Commitment anchoring at its own best fit: effects that were robust under offset anchoring and become indeterminate.

	observed	simulated	seed SD
R1, PMET duration	+0.04990	+0.00100	0.00115
R2, PMET coherence	+0.03060	+0.00036	0.00180
R12, delay duration	+0.03115	+0.00003	0.00078
R13, delay coherence	+0.00970	−0.00001	0.00090
R14, delay ramp	+0.00551	+0.00001	0.00009
R16, delay duration by time	+0.00139	−0.00003	0.00011

Where the problem comes from. A chain of three necessities, each forced by the previous one.

First, R9 forces $\alpha > \bar{v}$ by Proposition **??**. With $\alpha = 1.198$ the ramp adds $\alpha \times 3.22 = 3.86$ over the delay, far more than $g - \theta = 0.81$, so without a bound the level would cross the gate long before the response signal and every trial would respond prematurely. A saturation level is therefore forced.

Second, C must lie close to θ to hold premature responses down; the calibration puts $C - \theta = 0.30$.

Third, at $\alpha = 1.198$ the ramp covers that distance in $0.30/1.198 = 0.25$ seconds. Every trial therefore saturates within a quarter second of committing, and saturates at the *same* value C regardless of coherence and duration. The maintained state is uniform across all six cells for almost the whole delay, and every effect that depends on differences in the maintained level is annihilated. The delay-period total falls to 0.884 against 1.411 observed and premature responses to 0.03 per cent against 1.35 per cent.

There is no intermediate window. Lowering α restores the level differences but loses R9 and R6; raising it uniformises the state. The two requirements are monotone in α and opposite in direction.

Remark 9. This does not establish that commitment anchoring is wrong in general. The general argument for it is sound. What it establishes is that *in this design*, where the delay is fixed at 3 s and the stimulus lasts at most 2 s, the data require the offset. That is a falsifiable claim, and Section 8 gives the experiment that would test it.

6 R15

6.1 The contradiction, stated exactly

By Corollary **??**, R15 requires a substantial fraction of 2-second trials to fail to reach the internal threshold; otherwise the maintained level saturates at θ for every coherence and the coherence effect vanishes. The accuracy data forbid this. Sweeping θ with all other parameters at their calibrated values:

Table 4: R15 against accuracy, as a function of the internal threshold.

θ	R15	accuracy, 2 s, coh 0.12	accuracy, 2 s, coh 0.50
0.89 (calibrated)	−0.023	0.931	0.974
1.10	−0.025	0.963	0.993
1.30	+0.023	0.975	0.997
1.50	+0.092	0.972	0.998
1.70	+0.103	0.871	1.000
observed	+0.210	*0.789*	*0.952*

R15 turns positive only for θ between roughly 1.3 and 1.8, and never reaches half the observed value. At exactly those values, accuracy at 2 s and 12 per cent coherence rises to 0.97–0.98 against 0.789 observed, and at 50 per cent to 0.997–1.000 against 0.952.

These are the cells from which the manuscript draws its argument that accumulation terminates. The contradiction is therefore between R15 and that argument, not between R15 and an incidental feature of the fit.

6.2 Independent confirmation that accumulation terminates

AGPS (2025) Experiment 2 uses three stimulus durations rather than two, which localises the termination:

Table 5: Accuracy across three stimulus durations, AGPS 2025 Experiment 2.

coherence	300 ms	1000 ms	2000 ms	gain 300–1000	gain 1000–2000
2%	0.515	0.530	0.524	+0.015 ($p = .38$)	−0.006 ($p = .69$)
11%	0.724	0.774	0.777	+0.050 ($p = .002$)	+0.003 ($p = .81$)
40%	0.965	0.964	0.969	−0.001 ($p = .90$)	+0.005 ($p = .32$)

Unbounded accumulation predicts d' ratios of $\sqrt{1000/300} = 1.83$ and $\sqrt{2} = 1.41$; observed at mid coherence are 1.26 then 1.01. Accumulation uses the evidence up to roughly one second and then stops.

6.3 Nine mechanisms, none of which works

Each was implemented, calibrated where relevant, and evaluated on all nineteen effects.

mechanism	outcome
Accelerating preparation	Corrects the shape of the rate profile at constant α, but matching total preparation requires doubling α, which drives premature responses to 48–58 per cent and MET to 300–320 ms.
Force-integration response stage	Reduces premature responses eighteen-fold at matched MET, a real gain, but does not move R15. The physiologically faithful second-order twitch performs worse than a first-order leak, and worse still with a distributed motor-unit pool.
Strategic saturation near the gate	Compresses the most strongly prepared trials, which are the 2-second ones, reducing the coherence effect where R15 requires it increased.
Evidence-dependent maintenance noise	Calibration retains $\eta = 0.55$, so the direction is right, but it delivers 40 per cent of the required gradient while raising the loss from 6.6 to 19.6 and collapsing 2-second premature responses to 0.07–0.22 per cent.
Saturation level C	Invisible at a 3-second delay for $C > 1.33$, so it cannot affect R15 there. It is required for longer delays, for reasons independent of R15.
Leak inside the filter	Contracts differences with time constant $1/(\lambda + \mu) < 50$ ms, so it rescales once rather than eroding over seconds; and it does not bound a linearly growing input, so it does not replace C.
Leak on the maintained state	Decays toward zero rather than a non-zero endpoint. At $\mu = 0.25$ per second the delay-period rate collapses fourteen-fold.
Funnel toward $\pm\theta$	Uniformises the maintained state: R15 tends to zero without changing sign and R3 is extinguished with it.
Expansive peripheral projection	The only mechanism reproducing R15 at the right magnitude (+0.223 against +0.218 at $\gamma = 2.5$), at the cost of 81 per cent premature responses and an inverted R3. R15 and the premature rate are monotonically linked through the same parameter, so no window exists.

The pattern is the one Proposition **??** predicts: every mechanism that moves R15 acts through

the distance $g - m$, and therefore also moves the premature rate, the absolute rate, or premotor execution time.

6.4 How robust is R15 itself

The manuscript reports $F(2, 42) = 3.474$, $p = .040$, $\eta_p^2 = 0.142$. A planned contrast on the extreme coherence levels, computed per participant, gives +0.2104, $t(21) = 2.08$, $p = 0.0501$, $d = 0.44$; Wilcoxon gives $p = 0.0738$. Fourteen of twenty-two participants are positive, sign test $p = 0.29$. Under Benjamini–Hochberg correction across the nineteen effects, R15 does not survive. The between-participant standard deviation is 0.475, that is 2.3 times the mean; the bootstrap interval is $[+0.013, +0.401]$; leave-one-out moves p between 0.016 and 0.094.

At the subject level, the within-subject standard error of R15, obtained by bootstrapping trials within each participant, has median 0.425 against a between-subject standard deviation of 0.475. The ratio is 1.12: almost all of the apparent between-subject spread is sampling noise within participants, so there is very little genuine individual variability to explain. Compared against the model prediction of −0.026, three participants of twenty-two differ by more than two standard errors, all positive; roughly one would be expected by chance.

6.5 R15 in the companion experiment

AGPS (2025) Experiment 2 reports $F(1.86, 39.04) = 2.25$, $p = .12$, $\mathrm{BF}_{\mathrm{incl}} = 0.70$, anecdotal evidence for the null.

In matched units — partial activations per second, extreme coherence contrast — the two experiments give +0.105 per second ($p = 0.0501$, $d = 0.44$) in J Neuroscience and +0.138 per second ($p = 0.086$, $d = 0.38$) in AGPS. The effect is *larger* in the companion experiment. It replicates in direction and magnitude and fails to reach significance in either study alone. Stouffer combination gives $z = 2.74$, $p = 0.006$.

R15 should therefore not be treated as a false positive. It is a modest effect that neither experiment can establish alone and that the two together do establish. The model's failure is correspondingly substantive.

6.6 Critical assessment of the manuscript's treatment

Three explanations are offered, in two places, and they are mutually inconsistent.

Exclusion of anticipatory trials (p. 18, lines 350–359). The manuscript concedes it is insufficient. It can now be quantified: including premature trials with person-time denominators changes every reconstructed quantity in the third decimal place, since they are 1.35 per cent of trials and mostly occur late in the delay.

Strategic reduction of preparation near the gate (p. 18, lines 359–363). Implemented and tested, it acts in the wrong direction: it compresses differences among the most strongly prepared trials, which are the 2-second trials, reducing the coherence effect exactly where R15 needs it increased. R15 remains negative across the range of preparation slopes tested.

Differential threshold reaching (p. 22–23, lines 460–468). This is the second of the two branches set out at p. 9–10, lines 143–151. The Results invoke the *first* branch for R3 — "coherence-related differences in the starting point of the second accumulation phase were smaller in the 2 s condition" (p. 16, lines 313–315) — and the Discussion invokes the second for R15 — "increasing motion duration appears to have amplified coherence-related differences in sustained motor preparatory activity" (p. 23, lines 466–468). Both statements concern the same quantity, and the Methods state explicitly that the branches predict opposite interactions. Simulation confirms they are exclusive: the calibrated model implements the first branch, which R3 requires, and that forces the wrong sign for R15.

R24, the apparent levelling off before the response signal, is used to support the strategic-reduction account. It requires no mechanism: the rate is bounded by $\beta/2\pi$ and its sensitivity

$\mathrm{d}\nu/\mathrm{d}\mu = \nu a/\sigma_Y$ tends to zero as the level approaches the gate, so the highest-rate condition flattens first.

Recommendation. State R15 as a limitation with the contradiction of Section **??** made explicit, and withdraw all three proposed explanations. The contradiction between Results and Discussion should be resolved in one direction or the other regardless of what is done about R15.

7 R3

7.1 Status in the two experiments

J Neuroscience: $F(2, 42) = 3.62$, $p = .035$. AGPS (2025) Experiment 2, nearly the same design: $F(4, 84) < 1$, $\mathrm{BF}_{\mathrm{incl}} = 0.05$, that is *strong* evidence for the null.

The discrepancy is not a matter of the coherence range at the low end. In J Neuroscience the interaction is carried entirely by the 12 versus 50 per cent contrast: dropping the zero-coherence condition *strengthens* it, from +0.0179 ($p = 0.075$) to +0.0210 ($p = 0.0044$), while the 0 versus 12 per cent contrast gives −0.0032 ($p = 0.73$).

7.2 Where the difference lies

Simple coherence effects on premotor execution time, mid versus high coherence:

	short stimulus	long stimulus
J Neuroscience (12 vs 50%)	+28.2 ms ($p = .007$)	+7.1 ms ($p = .15$)
AGPS Exp. 2 (11 vs 40%)	+30.5 ms	+27.2 ms

The short-duration effect replicates almost exactly, 28.2 against 30.5 ms. The entire discrepancy is at long duration: coherence ceases to influence premotor execution time after 2 s in J Neuroscience and continues to do so in AGPS.

7.3 The explanation I proposed, and why it is refuted

The natural explanation is that accumulation terminates before 2 s in one experiment and not the other, so the maintained level saturates at θ across coherences in one and not the other. This is testable through accuracy, since termination implies no further accuracy gain.

It is refuted. The accuracy gains from short to long stimulus are essentially identical: at mid coherence +0.075 ($p = .0009$) in J Neuroscience against +0.053 ($p = .0146$) in AGPS; at high coherence +0.000 against +0.004. The degree of saturation of the decision process is the same in the two experiments.

7.4 What this leaves

At equal decision quality, coherence still influences premotor execution time after a 2-second stimulus in one experiment and not in the other. The difference must therefore lie between the decision and premotor execution time, not in the accumulation. I have no explanation.

Two factual differences are on record and neither is obviously sufficient. Stimulus parameters differ: dot diameter 0.05° against 0.06°, dot speed 10.5 against 6.47 degrees per second, and different aperture sizes. And the coherence levels differ at both ends, 0 and 50 per cent against 2 and 40 per cent.

7.5 Critical assessment of the manuscript's treatment

The Methods, p. 9–10, lines 136–156, are careful and correct: the temporal preparation hypothesis does not predict the interaction; the threshold and combined hypotheses allow it; and its *direction* is explicitly not predicted, since "the direction of this interaction depends on how the additional motion evidence provided by the 2 s duration changes the relative probability of reaching the internal threshold across coherence levels".

That clause has a consequence the manuscript does not draw. If both directions are permitted in advance, observing one of them discriminates against the temporal preparation hypothesis but provides no test of the combined hypothesis. R3 should be presented as evidence against H2, not as evidence for H3.

The account given in the Results also contradicts the account given in the Discussion for R15, as documented in Section 6.

On robustness. Given the failure to replicate in the companion experiment, and given that R3 does not survive multiple-comparison correction, it should be reported with that context. The model reproduces it (+0.0349 simulated against +0.0179 observed, inside the observed confidence interval), but reproducing an effect that does not replicate is weak support.

8 A thought experiment that discriminates the anchorings

Consider two conditions with the same total time from stimulus onset to response signal: condition 1 with a 5-second stimulus and a 5-second delay, condition 2 with an 8-second stimulus and a 2-second delay.

Both stimuli are long enough that essentially every trajectory reaches θ, so the maintained level and the accuracy are the same. The only quantity that differs is the duration of motor preparation.

Proposition 19. *Under offset anchoring, preparation runs for the length of the delay, 5 s against 2 s, so the level at the response signal is higher in condition 1 by 3α: premotor execution time is shorter and the delay-period rate higher in condition 1. Under commitment anchoring, preparation runs from T_{stop} to the response signal, $10 - T_{stop}$ seconds in both, so the two conditions are predicted to be identical.*

This requires neither electroencephalography nor measured excursion durations. It is not realisable in the existing data: no pair of conditions in the three experiments has matched total time with a different split.

The partial equivalent in the present data. R1 decomposes differently under the two anchorings. Under offset anchoring the ramp lasts 3 s in both duration conditions, so the entire 49.9 ms effect must come from differences in m_0. Under commitment anchoring the ramp lasts 1.71 s longer after a 2-second stimulus, contributing $\alpha \times 1.71/w \approx 37$ ms at the offset-calibrated α, so most of the effect would come from the ramp. Both variants can produce a 50 ms effect, so R1 does not discriminate by itself. What discriminates in the present data is R9 and R6, through the plateau.

An accessible version. Crossing a short delay with a long stimulus — 2 s of stimulus with a 1 s delay against 0.3 s of stimulus with a 1 s delay — separates the two without requiring ten-second trials. Under offset anchoring the ramp is 1 s in both and R9 survives; under commitment anchoring the long-stimulus condition prepares throughout the stimulus and R9 should reverse.

9 Is information after the internal threshold discarded?

The model assumes evidence arriving after T_{stop} is discarded entirely. The three stimulus durations of AGPS Experiment 2 (Table 5) support the assumption and make it more precise than the manuscript can: accumulation uses the evidence up to roughly one second at intermediate coherence and earlier at high coherence, after which further evidence does not contribute to the decision. The second half of a 2-second stimulus contributes nothing.

One distinction should be drawn that the manuscript does not. What the accuracy data establish is that the *choice* is settled, since accuracy depends only on the sign. They place no constraint on whether the *level* continues to be updated after commitment. The manuscript treats these as one claim at p. 22, lines 439–443. Separating them is what the title of the paper implies: a decision that is locked while the motor level remains graded.

10 Summary

- With preparation added outside the filter and anchored to stimulus offset, sixteen of nineteen effects are reproduced robustly, including R9 to within four per cent.
- Commitment anchoring, the more natural specification and the one originally proposed, cannot be made to work in this design. R9 forces a nine-fold increase in the preparation slope; that forces a saturation level; and the saturation uniformises the maintained state, annihilating six effects the offset variant reproduces.
- R15 fails, and Proposition **??** shows why: it requires 2-second trials not to reach the internal threshold, which the accuracy data forbid. Nine mechanisms were tested and none removes the contradiction.
- R15 is nonetheless a real effect: it replicates in the companion experiment with a larger effect size, and the two combined give $p = 0.006$.
- R3 is reproduced here but does not replicate in the companion experiment. The difference lies entirely at long stimulus duration and is not explained by differential termination of accumulation, since the accuracy gains are the same in the two experiments.
- The model is uniformly too reactive to condition: seven effects exceed the observed confidence interval and none falls below. This may be the same deficiency as the 45 per cent underestimation of trial-to-trial variability in premotor execution time.